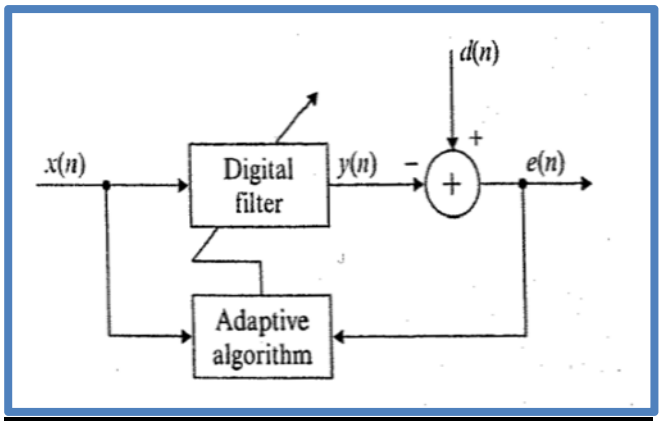
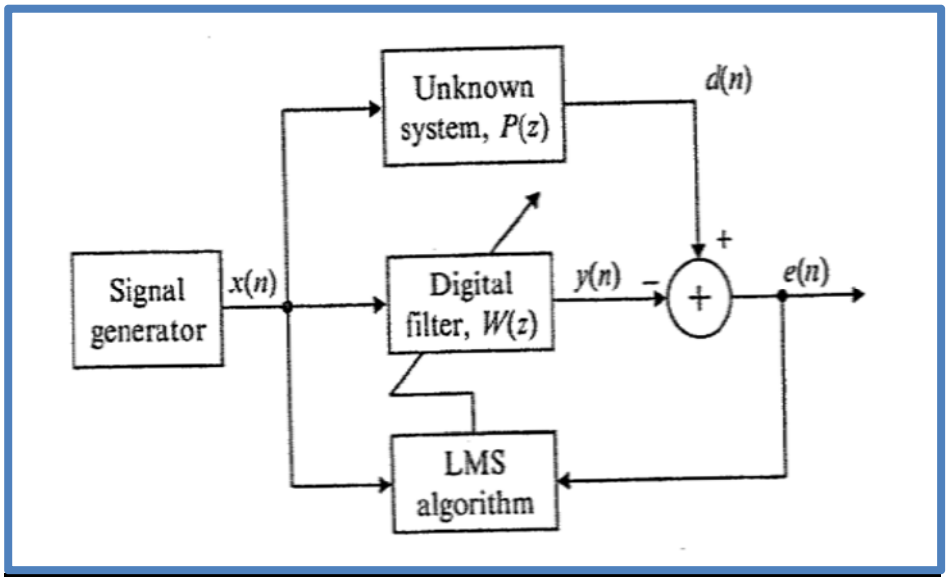
Generic applications of adaptive filters

General model



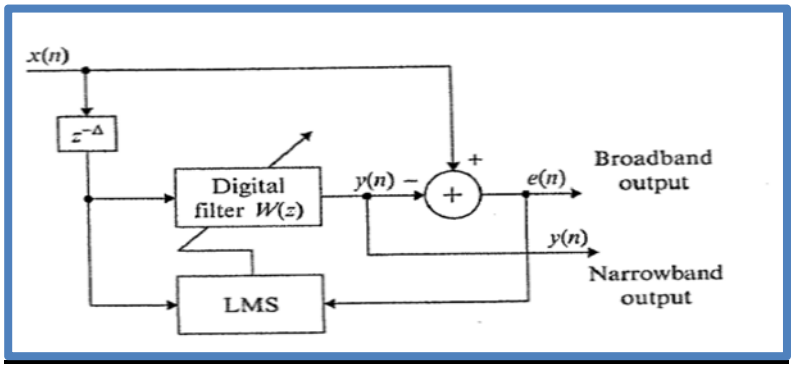
Adaptive System Identification



Domain:
Acoustics, Vibration, Communication, Control, …

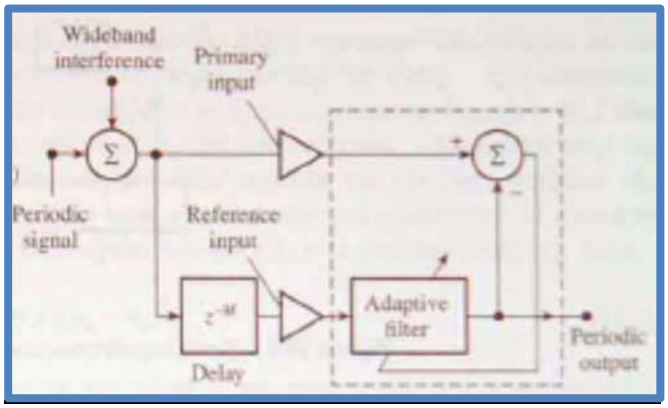
Operation:
Adaptive algorithm minimizes $\mathbf{e(n)} \Rightarrow \mathbf{W(z)} \cong \mathbf{P(z)}$

Adaptive Linear Prediction (aka. ALE, Adaptive Line Enhancement)



Domain:
Communications, …

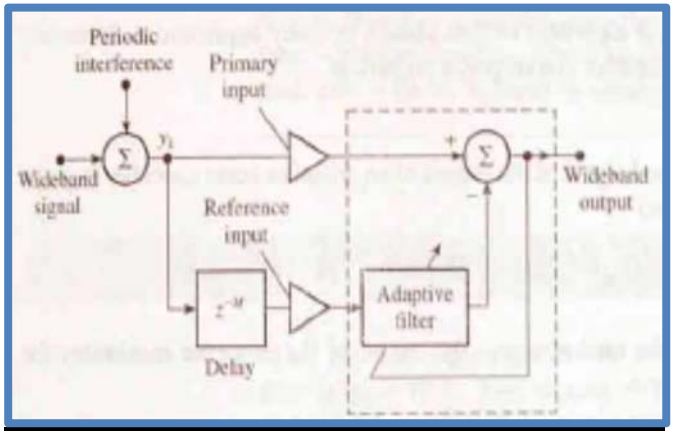
Operation:
Signal $\mathbf{x(n) = s(n) + v(n)}$ consists of narrowband part, $\mathbf{s(n)}$, and wideband part, $\mathbf{v(n)}$
Adaptive algorithm minimizes $\mathbf{e(n)}$ by linear prediction of $\mathbf{x(n)}$ from $\mathbf{x(n-\Delta)}$
The delay, Δ, is chosen big enough to ensure $\mathbf{R_{vv}(k) \cong 0}$ for $\mathbf{|k| \geq \Delta}$ (e.g. WN: $\Delta = 1$)
Then $\mathbf{v(n)}$ is uncorrelated with $\mathbf{v(n-\Delta)}$, but $\mathbf{s(n)}$ is correlated with $\mathbf{s(n-\Delta)}$
$\Rightarrow$ $\mathbf{y(n) \cong s(n)}$ and $\mathbf{e(n) \cong v(n)}$



Mode 1:
Classical noise reduction

Narrowband signal s(n)
Wideband (e.g. white) noise v(n)

y(n) will be noise-reduced signal
e(n) will be noise part

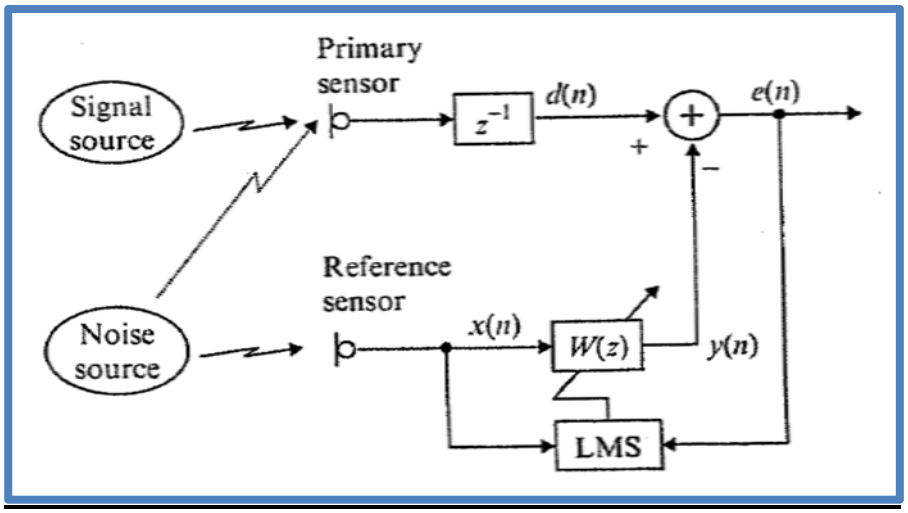


Mode 2:
3G/4G wireless communication

Because of spread-spectrum coding of a narrowband signal, it will appear wideband (noise-like), thus acting as v(n)
s(n) might then consist of narrowband interference (e.g. from power-line or jamming)

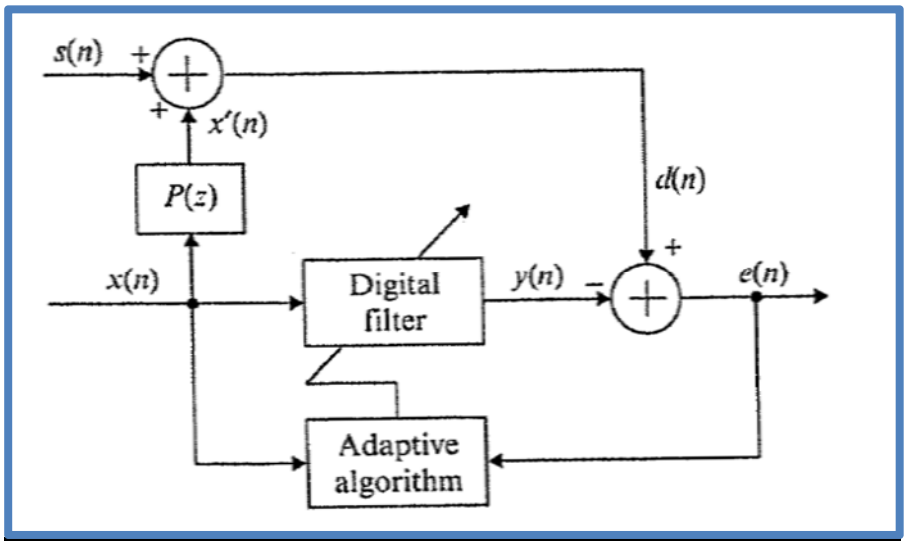
e(n) will be the wideband signal
y(n) will be the interference

Adaptive (Active) Noise Cancellation, ANC

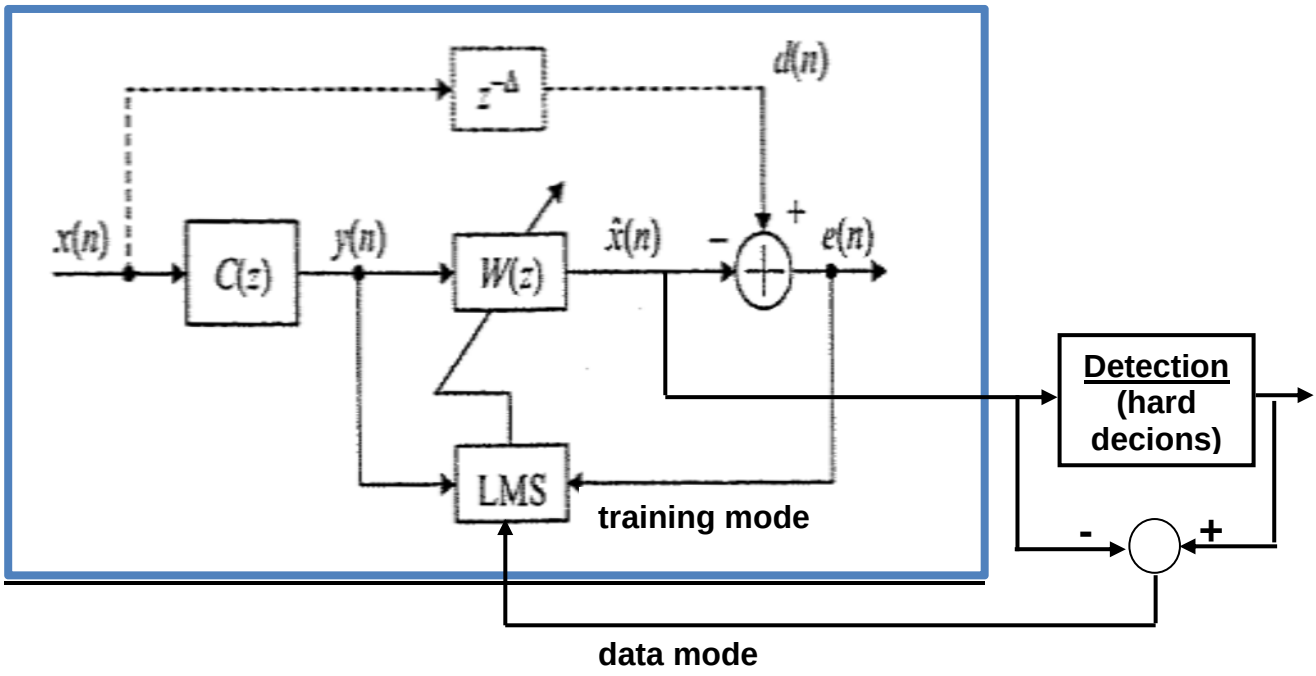


Domain:
Acoustics, Vibration, …

Operation:
This is (unlike ALE) a 2-channel system
Primary sensor (e.g. MIC) is close to the signal source but also has noise parts
Secondary sensor is close to the noise source (measures a reference to the noise)
The noise components in the 2 sensors are not identical due to (e.g. acoustic) transfer function from noise source to sensors, but are correlated
The adaptive algorithm minimizes $\mathbf{e(n) = d(n) - y(n) = s(n) + x'(n) - y(n)}$
Now $\mathbf{E[e^2(n)] = E[s^2(n)] + E[(x'(n) - y(n))^2] + 2E[s(n)x'(n)] - 2E[s(n)y(n)]}$
$\mathbf{= E[s^2(n)] + E[(x'(n) - y(n))^2]}$
Minimization: $\mathbf{y(n) = x'(n) \Rightarrow W(z) = P(z) \Rightarrow e(n) = s(n)}$ **(note that here** $\mathbf{e(n) \neq 0}$**)**



Adaptive Inverse Modelling



Domain:
Digital communication, …

Operation:
$\mathbf{C}(\mathbf{z})$ models undesired (linear, but possibly time-varying) distortion on a communications transmission channel from sender to receiver
The adaptive filter, $\mathbf{W}(\mathbf{z})$, is compensating for the distortion, ideally $\mathbf{C}(\mathbf{z})\mathbf{W}(\mathbf{z}) = \mathbf{1}$
Practically, there is a delay, Δ, through channel and filter so that $\mathbf{C}(\mathbf{z})\mathbf{W}(\mathbf{z}) = \mathbf{z}^{-\Delta}$
In initial training mode a set of known signals (symbols) are sent through the channel according to a communications protocol
This adjusts the adaptive filter, $\mathbf{W}(\mathbf{z})$
Then the protocol switches to data mode where unknown signals (symbols) are sent over the channel
In the data mode $\mathbf{W}(\mathbf{z})$ can then adapt to possible changes on the characteristics of the communication channel using "decision-directed updating"
If the SNR is reasonably high, this is done by assuming that the hard-decisions (detection) of received symbols are correct and then use the difference between the actually received symbol values and the detected values to update the adaptive filter